

Math 1522 Calculus 2 Review: Selected Solutions
Fall 2026

1. Formulas to know

- (b) State the Inverse Function Theorem.

Solution. Let f be a 1-1 differentiable function with inverse f^{-1} , and let a be in the domain of f^{-1} . If $f'(f^{-1}(a)) \neq 0$, then f^{-1} is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

The tangent line to f at (b, a) , where $b = f^{-1}(a)$, reflected across $y = x$ is the tangent line to f^{-1} at (a, b) , and reflecting swaps rise and run. ■

2. Logarithmic differentiation

- (c) Find $y'(0)$ if $y = \frac{(x+1)^4 e^x}{\sqrt{1+3x}}$.

Solution. Take $\ln$ of both sides and expand: $\ln y = 4 \ln(x+1) + x - \frac{1}{2} \ln(1+3x)$. Differentiate implicitly.

$$\frac{y'}{y} = \frac{4}{x+1} + 1 - \frac{3}{2(1+3x)}.$$

At $x = 0$, $y = 1$, so $y'(0) = 4 + 1 - \frac{3}{2} = \frac{7}{2}$. ■

3. Derivatives of logarithms

- (c) Let $f(x) = 3 - e^{-x}$. Find the domain and range of f , find $f^{-1}(x)$, and state the domain and range of f^{-1} . Does the graph of f have a horizontal asymptote?

Solution. f is defined for every x , so the domain is $(-\infty, \infty)$. The values of e^{-x} fill $(0, \infty)$, so the range of f is $(-\infty, 3)$.

Inverse: set $y = 3 - e^{-x}$ and solve for x . Then $e^{-x} = 3 - y$, so $x = -\ln(3 - y)$. Swapping names,

$$f^{-1}(x) = -\ln(3 - x),$$

with domain $(-\infty, 3)$ (the range of f) and range $(-\infty, \infty)$ (the domain of f).

As $x \rightarrow \infty$, $e^{-x} \rightarrow 0$ so $f(x) \rightarrow 3$. The line $y = 3$ is a horizontal asymptote. ■

4. Inverse trig functions

- (c) Find the equation of the tangent line to $y = 2 \tan^{-1}(x^2)$ at $x = 1$.

Solution. The point is $y(1) = 2 \tan^{-1}(1) = \frac{\pi}{2}$. The slope is

$$y' = 2 \cdot \frac{2x}{1+x^4}, \quad y'(1) = 2.$$

So the tangent line is $y = \frac{\pi}{2} + 2(x - 1)$. ■

5. Hyperbolic functions

(a) Find $\cosh^{-1}(2)$.

Solution. Solve $\cosh y = 2$, that is $\frac{e^y + e^{-y}}{2} = 2$. Multiply by $2e^y$: $e^{2y} - 4e^y + 1 = 0$, a quadratic in e^y . So $e^y = 2 \pm \sqrt{3}$ and $y = \ln(2 \pm \sqrt{3})$. Since $(2 - \sqrt{3})(2 + \sqrt{3}) = 1$, the two solutions are $y = \pm \ln(2 + \sqrt{3})$. The function $\cosh^{-1}$ is the inverse of $\cosh$ restricted to $[0, \infty)$, so

$$\cosh^{-1}(2) = \ln(2 + \sqrt{3}).$$

■

6. Inverse Function Theorem

(c) Let $f(x) = x + e^x$. Find the equation of the tangent line to the graph of f^{-1} at $x = 1$.

Solution. $f(0) = 0 + 1 = 1$, so $f^{-1}(1) = 0$ and the point is $(1, 0)$. $f'(x) = 1 + e^x$, so $(f^{-1})'(1) = \frac{1}{f'(0)} = \frac{1}{2}$. The tangent line is

$$y = \frac{1}{2}(x - 1).$$

■

7. L'Hôpital's rule

(b) $\lim_{x \rightarrow 0^+} (1 + \sin(2x))^{\cot x}$

Solution. The form 1^∞ . Let y be the expression, so

$$\ln y = \cot x \ln(1 + \sin 2x) = \frac{\ln(1 + \sin 2x)}{\tan x}, \text{ the form } \frac{0}{0}.$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 2x)}{\tan x} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{\frac{2 \cos 2x}{1 + \sin 2x}}{\sec^2 x} = \frac{2}{1} = 2.$$

So the limit is e^2 .

■

8. Substitution

(e) $\int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$

Solution. Let $u = 1 + \sqrt{x}$, $du = \frac{dx}{2\sqrt{x}}$.

$$\int \frac{2 du}{u} = 2 \ln |u| + C = 2 \ln(1 + \sqrt{x}) + C.$$

■

9. Integration by parts

(b) $\int_0^\pi e^x \sin(x) dx$

Solution. Call the antiderivative $I = \int e^x \sin x \, dx$. Parts with $u = \sin x$, $dv = e^x \, dx$, then again with $u = \cos x$, $dv = e^x \, dx$:

$$I = e^x \sin x - \int e^x \cos x \, dx = e^x \sin x - e^x \cos x - I.$$

So $I = \frac{e^x(\sin x - \cos x)}{2} + C$, and

$$\int_0^\pi e^x \sin x \, dx = \frac{e^\pi(0+1)}{2} - \frac{1 \cdot (0-1)}{2} = \frac{e^\pi + 1}{2}.$$

10. Trig integrals

(c) $\int \tan^2(x) \sec^4(x) \, dx$

Solution. The power of $\sec$ is even, so save $\sec^2 x$ and convert the other $\sec^2 x = 1 + \tan^2 x$. Let $u = \tan x$, $du = \sec^2 x \, dx$.

$$\int \tan^2 x (1 + \tan^2 x) \sec^2 x \, dx = \int (u^2 + u^4) \, du = \frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C.$$

11. Trig substitution

(a) $\int_2^3 \frac{dx}{(x^2-1)^{3/2}}$

Solution. Let $x = \sec \theta$, $dx = \sec \theta \tan \theta \, d\theta$, so $(x^2 - 1)^{3/2} = \tan^3 \theta$ (for $0 \leq \theta < \frac{\pi}{2}$).

$$\int \frac{\sec \theta \tan \theta}{\tan^3 \theta} \, d\theta = \int \frac{\cos \theta}{\sin^2 \theta} \, d\theta = -\frac{1}{\sin \theta} + C = -\frac{x}{\sqrt{x^2-1}} + C,$$

since $\sin \theta = \frac{\sqrt{x^2-1}}{x}$ from the triangle with hypotenuse x and adjacent side 1.

$$\int_2^3 \frac{dx}{(x^2-1)^{3/2}} = -\frac{3}{\sqrt{8}} + \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3} - \frac{3\sqrt{2}}{4}.$$

12. Partial fractions

(c) $\int \frac{3x^2-x+8}{x^3+4x} \, dx$

Solution. $x^3 + 4x = x(x^2 + 4)$ with $x^2 + 4$ irreducible. Write $\frac{3x^2-x+8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+D}{x^2+4}$, so $3x^2 - x + 8 = A(x^2 + 4) + (Bx + D)x$. At $x = 0$, $A = 2$. Matching x^2 : $3 = A + B$, so $B = 1$. Matching x : $D = -1$.

$$\int \left(\frac{2}{x} + \frac{x}{x^2+4} - \frac{1}{x^2+4} \right) dx = 2 \ln |x| + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C.$$

13. Improper integrals

(d) $\int_0^\infty \frac{e^x}{(1+e^x)^3} dx$

Solution. Let $u = 1 + e^x$, $du = e^x dx$. Then $x = 0$ gives $u = 2$ and $x \rightarrow \infty$ gives $u \rightarrow \infty$.

$$\lim_{t \rightarrow \infty} \int_2^t u^{-3} du = \lim_{t \rightarrow \infty} \left(\frac{1}{2 \cdot 4} - \frac{1}{2t^2} \right) = \frac{1}{8}.$$

■

14. Numerical integration

(a) $\int_0^1 (x^3 - 3x^2 + 2x + 1) dx$: over or under estimates, then M_4 and T_4 .

Solution. Let $f(x) = x^3 - 3x^2 + 2x + 1$. Then $f''(x) = 6x - 6 < 0$ on $[0, 1]$, so f is concave down. A concave down graph lies above its chords and below its tangent lines, so the trapezoid rule underestimates and the midpoint rule overestimates.

$f'(x) = 3x^2 - 6x + 2$ changes sign at $x = 1 - \frac{1}{\sqrt{3}}$, so f is not monotonic on $[0, 1]$, and the left and right endpoint rules are not determined by this reasoning.

With $\Delta x = \frac{1}{4}$, $M_4 = \frac{1}{4} \left(f\left(\frac{1}{8}\right) + f\left(\frac{3}{8}\right) + f\left(\frac{5}{8}\right) + f\left(\frac{7}{8}\right) \right) = \frac{161}{128} \approx 1.258$ and $T_4 = \frac{1}{8} \left(f(0) + 2f\left(\frac{1}{4}\right) + 2f\left(\frac{1}{2}\right) + 2f\left(\frac{3}{4}\right) + f(1) \right) = \frac{79}{64} \approx 1.234$. The exact value is $\frac{1}{4} - 1 + 1 + 1 = \frac{5}{4}$, between them as predicted.

■

15. Sequences

(d) Show that $\lim_{n \rightarrow \infty} \frac{3^n}{n!} = 0$.

Solution. For $n \geq 3$, write the quotient as a product of n factors and bound the middle ones by 1:

$$0 < \frac{3^n}{n!} = \frac{3}{1} \cdot \frac{3}{2} \cdot \frac{3}{3} \cdot \frac{3}{4} \cdots \frac{3}{n-1} \cdot \frac{3}{n} \leq \frac{3}{1} \cdot \frac{3}{2} \cdot \frac{3}{n} = \frac{27}{2n}.$$

Since $\frac{27}{2n} \rightarrow 0$, the Squeeze Theorem gives $\frac{3^n}{n!} \rightarrow 0$.

■

16. Test for Divergence (TFD)

(e) Suppose $b_n > 0$ and $\sum b_n$ converges. What does the TFD say about $\sum \frac{1}{b_n}$?

Solution. Since $\sum b_n$ converges, $b_n \rightarrow 0$. As $b_n > 0$, this means $\frac{1}{b_n} \rightarrow \infty$, so the terms of $\sum \frac{1}{b_n}$ do not go to 0. By the TFD, $\sum \frac{1}{b_n}$ diverges.

■

17. Telescoping series

(a) $\sum_{n=2}^\infty \frac{1}{n^2-1}$

Solution. Partial fractions: $\frac{1}{n^2-1} = \frac{1}{2} \left(\frac{1}{n-1} - \frac{1}{n+1} \right)$. In the partial sum $s_N = \sum_{n=2}^N$, everything cancels except the first two positive terms and the last two negative ones.

$$s_N = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N} - \frac{1}{N+1} \right) \rightarrow \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}.$$

■

18. Geometric series

- (e) For which a does $\sum_{k=1}^{\infty} \left(\frac{2a}{1+a^2}\right)^k$ converge? Find its value.

Solution. Let $r = \frac{2a}{1+a^2}$. The series converges exactly when $|r| < 1$. Since $(|a| - 1)^2 \geq 0$, we get $2|a| \leq 1 + a^2$, so $|r| \leq 1$, with equality exactly when $|a| = 1$. So the series converges for all $a \neq \pm 1$. Its first term is r , so

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r} = \frac{2a}{1+a^2-2a} = \frac{2a}{(a-1)^2}.$$

■

19. Integral Test

- (a) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

Solution. $f(x) = \frac{1}{x(\ln x)^2}$ is positive, continuous, and decreasing on $[2, \infty)$. With $u = \ln x$,

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \lim_{t \rightarrow \infty} \left(\frac{1}{\ln 2} - \frac{1}{\ln t} \right) = \frac{1}{\ln 2}.$$

The integral converges, so the series converges.

■

20. p-series

- (d) For which p does $\sum_{n=1}^{\infty} \frac{1}{n^{2p-1}}$ converge?

Solution. It is a p -series with exponent $2p - 1$, so it converges exactly when $2p - 1 > 1$, that is $p > 1$.

■

21. Direct Comparison Test (DCT) and Limit Comparison Test (LCT)

- (a) $\sum_{n=1}^{\infty} \sin(1/n)$

Solution. The terms are positive. Compare with $b_n = \frac{1}{n}$, using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$:

$$\lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = 1.$$

The limit is positive and finite and $\sum \frac{1}{n}$ diverges, so by the LCT the series diverges.

■

22. Alternating Series Test (AST)

- (e) $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n}}{n+4}$

Solution. Let $b_n = \frac{\sqrt{n}}{n+4}$ and $f(x) = \frac{\sqrt{x}}{x+4}$. Then

$$f'(x) = \frac{\frac{x+4}{2\sqrt{x}} - \sqrt{x}}{(x+4)^2} = \frac{4-x}{2\sqrt{x}(x+4)^2} < 0 \quad \text{for } x > 4,$$

so b_n is decreasing for $n \geq 4$. Dividing top and bottom by $\sqrt{n}$, $b_n = \frac{1}{\sqrt{n+4}/\sqrt{n}} \rightarrow 0$. The AST applies from $n = 4$ on, so the series converges.

■

23. Ratio Test

(c) $\sum_{n=1}^{\infty} \frac{n^2 4^n}{n!}$

Solution.

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2 4^{n+1}}{(n+1)!} \cdot \frac{n!}{n^2 4^n} = \frac{4(n+1)}{n^2} \rightarrow 0 < 1.$$

By the Ratio Test it converges. ■

24. Root Test

(a) $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1} \right)^n$

Solution.

$$\sqrt[n]{a_n} = \frac{2n+1}{3n-1} \rightarrow \frac{2}{3} < 1.$$

By the Root Test it converges. ■

25. Absolute and conditional convergence

(d) $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$

Solution. Absolute values: $\frac{n}{n^2+1}$ compared with $\frac{1}{n}$ gives $\lim \frac{n^2}{n^2+1} = 1$, so $\sum \frac{n}{n^2+1}$ diverges by the LCT. AST: $b_n = \frac{n}{n^2+1} \rightarrow 0$, and $f(x) = \frac{x}{x^2+1}$ has $f'(x) = \frac{1-x^2}{(x^2+1)^2} \leq 0$ for $x \geq 1$, so b_n is decreasing. The series converges conditionally. ■

26. Radius of Convergence (ROC) and Interval of Convergence (IOC)

(c) $\sum_{n=1}^{\infty} \frac{(x+2)^n}{n 4^n}$

Solution. Ratio Test: $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x+2|}{4} \cdot \frac{n}{n+1} \rightarrow \frac{|x+2|}{4}$. So it converges for $|x+2| < 4$: center -2 , $R = 4$. At $x = 2$: $\sum \frac{1}{n}$ diverges. At $x = -6$: $\sum \frac{(-1)^n}{n}$ converges by the AST. IOC $[-6, 2)$. ■

27. Power series from geometric series

(e) Find the exact value of $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

Solution. Differentiate the geometric series and multiply by x : $\sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$ for $|x| < 1$. At $x = \frac{1}{3}$:

$$\sum_{n=1}^{\infty} \frac{n}{3^n} = \frac{1/3}{(2/3)^2} = \frac{3}{4}.$$
■

28. Taylor and Maclaurin series

(d) Maclaurin series for $F(x) = \int_0^x e^{-t^2} dt$ and its ROC.

Solution. Substitute $-t^2$ into $e^u = \sum \frac{u^n}{n!}$, then integrate term by term.

$$e^{-t^2} = \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{n!}, \quad F(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n! (2n+1)} = x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots$$

The series for e^u has $R = \infty$, and integrating does not change the radius, so $R = \infty$. ■

29. Approximations and error bounds

- (b) Use the degree 3 Maclaurin polynomial $T_3(x)$ to approximate $\tan^{-1}(x)$ near 0, and bound the error at $x = \frac{1}{2}$ with the Alternating Series Estimation Theorem (ASE).

Solution. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$, so $T_3(x) = x - \frac{x^3}{3}$ and $T_3(\frac{1}{2}) = \frac{1}{2} - \frac{1}{24} = \frac{11}{24}$. At $x = \frac{1}{2}$ the series alternates with decreasing terms, so the error is at most the next term:

$$|\tan^{-1}(\frac{1}{2}) - T_3(\frac{1}{2})| \leq \frac{(1/2)^5}{5} = \frac{1}{160}.$$

■

30. Complex numbers

- (a) Compute $(1+i)^{10}$ in $a+bi$ form.

Solution. $1+i = \sqrt{2} e^{i\pi/4}$. So $(1+i)^{10} = (\sqrt{2})^{10} e^{i10\pi/4} = 32 e^{i5\pi/2} = 32 e^{i\pi/2} = 32i$. ■

31. Differential equations

- (j) Logistic: $\frac{dP}{dt} = 0.5P \left(1 - \frac{P}{1000}\right)$, $P(0) = 200$. Carrying capacity, equilibria, growth, and $P(t)$.

Solution. (i) The carrying capacity is $M = 1000$. The equilibria are $P = 0$ and $P = 1000$.

(ii) For $0 < P < 1000$ both factors are positive, so P increases. The rate $0.5P - \frac{0.5P^2}{1000}$ is a downward parabola in P with roots 0 and 1000, so it is largest at the vertex $P = 500$.

(iii) Separate with partial fractions: $\frac{1000}{P(1000-P)} = \frac{1}{P} + \frac{1}{1000-P}$, so $\ln \left| \frac{P}{1000-P} \right| = 0.5t + C$ and $\frac{P}{1000-P} = Be^{0.5t}$ for a constant B . At $t = 0$, $B = \frac{200}{800} = \frac{1}{4}$. Solving for P ,

$$P(t) = \frac{1000}{1 + 4e^{-0.5t}}.$$

$P = 500$ when $4e^{-0.5t} = 1$, that is $t = 2 \ln 4 = 4 \ln 2$. ■